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Cohomology of Jordan Algebras

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INTRODUCTION

Let $\mathscr{J}$ be a Jordan algebra of degree larger than 3, $U(\mathscr{J})$ the universal enveloping algebra of $\mathscr{J}$, and M a Jordan bimodule for it. In Glassman [26], we defined an inner derivation functor to be any epimorphism-preserving subfunctor of the derivation functor, D. For any inner derivation functor J, and any short exact sequence $0 \to M' \to M \to M'' \to 0$, we showed the existence of a long exact sequence

$$\begin{aligned} 0 \to H_J^0(\mathscr{J}, M') &\to H_J^0(\mathscr{J}, M) \to H_J^0(\mathscr{J}, M'') \\ &\xrightarrow{\delta^0} H_J^1(\mathscr{J}, M') \to H_J^1(\mathscr{J}, M) \to H_J^1(\mathscr{J}, M'') \\ &\xrightarrow{\delta^1} H^2(\mathscr{J}, M') \to \cdots \to, \end{aligned}$$

where

$$H_J^1(\mathscr{J}, M) \simeq \frac{D(\mathscr{J}, M)}{J(\mathscr{J}, M)},$$

$H^n(\mathscr{J}, M)$ is isomorphic to the module of short singular extensions of length n of M by $\mathscr{J}$, and $H_J^0(\mathscr{J}, M)$ is isomorphic to $\mathrm{Hom}_{U(\mathscr{J})}(C, M)$ for some $\mathscr{J}$ bimodule C.

In this paper we show that $H^2(\mathscr{J}, M) = 0$ for M injective and determine an inner derivation functor $J(\mathscr{J}, -)$ such that $H_J^1(\mathscr{J}, M) = 0$ for M injective. Finally we give a simple form for the bimodule C such that

$$H_J^0(\mathscr{J}, M) = \mathrm{Hom}_{U(\mathscr{J})}(C, M)$$

and $H^n(\mathscr{J}, -)$ is the n-th right derived functor of $\mathrm{Hom}_{U(\mathscr{J})}(C, -)$. The advantage of the added generality of inner derivation functors as defined in Glassman [26] will become apparent since we will use one that is neither minimal, maximal, nor classical.

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Before beginning, we need some preliminary results concerning Jordan algebras and related constructions.

Koecher's Embedding

Let $\mathcal{J}$ be a finite dimensional Jordan algebra with unit over a field of characteristic $\neq 2,3$. Koecher [18] has associated with such an algebra a Lie algebra, $K_s(\mathcal{J})$ which closely reflects the structure of $\mathcal{J}$.

Write R_a for right multiplication by a in $\mathcal{J}$,

$$R(\mathcal{J}) = R_{\mathcal{J}} = \{R_a \mid a \in \mathcal{J}\}.$$

Let $\mathcal{L}(\mathcal{J})$ be the Lie transformation algebra of $\mathcal{J}$. The identity

$$[R_a , [R_b , R_c]] = R_{a[R_b, R_c]}$$

shows that

$$\mathcal{L}(\mathcal{J}) = R(\mathcal{J}) + [R(\mathcal{J}), R(\mathcal{J})].$$

Define $\Delta : \mathcal{J} \otimes \mathcal{J} \to \mathcal{L}(\mathcal{J})$ by

$$a\Delta b = R_{a.b} - [R_a , R_b] = R_a^{(b)}.$$

Note here, that multiplication in the Jordan algebra is denoted by dot product.

As a vector space,

$$K_s(\mathcal{J}) = \mathcal{J} \oplus \mathcal{J}' \oplus \mathcal{L}(\mathcal{J}),$$

where $\mathcal{J}'$ is linearly isomorphic to $\mathcal{J}$.

Define a vector space automorphism $'$ on $K_s(\mathcal{J})$ as follows:

$'$ exchanges components $\mathcal{J}$ and $\mathcal{J}'$, $R_a' = -R_a'$, $[R_a , R_b]' = [R_a , R_b]$.

We also write $\pi : K_s(\mathcal{J}) \to K_s(\mathcal{J})$ for this mapping.

Multiplication in $K_s(\mathcal{J})$ is given by

$$\begin{aligned}[a_1 + b_1' + l_1 , a_2 + b_2' + l_2] \\ = (a_1 l_2 - a_2 l_1) + ((b_1 l_2')' - (b_2 l_1')') + (a_1 \Delta b_2 - a_2 \Delta b_1 + [l_1 , l_2]).\end{aligned}$$

Here $l \in \mathcal{L}(\mathcal{J})$. It is an easy verification that $K_s(\mathcal{J})$ under this multiplication is a Lie algebra and π is an algebra automorphism of period 2. We also note that the subalgebra of $K_s(\mathcal{J})$ generated by 1, 1', R_1 is a Lie algebra of type A_1. Koecher has shown that $\mathcal{J}$ is semisimple (separable) if and only if $K_s(\mathcal{J})$ is semisimple (separable).

Now let M be a Jordan bimodule for $\mathscr{J}$. Then $E(\mathscr{J}, M)$, the split null extension of M by $\mathscr{J}$, is a Jordan algebra and we may construct

$$K_s(E(\mathscr{J}, M)) = (\mathscr{J} \oplus M) \oplus (\mathscr{J}' \oplus M') \oplus R_{\mathscr{J}} \oplus R_M \oplus [R_{\mathscr{J}}, R_{\mathscr{J}}] \oplus [R_{\mathscr{J}}, R_M].$$

Writing

$$i_s(M) = M \oplus M' \oplus R_M \oplus [R_{\mathscr{J}}, R_M],$$

we see that

$$K_s(E(\mathscr{J}, M)) = K_s(\mathscr{J}) \oplus i_s(M) = E(K_s(\mathscr{J}), i_s(M))$$

is a Lie algebra and so $i_s(M)$ is a Lie bimodule for $K_s(\mathscr{J})$.

We next examine the derivations of $K_s(\mathscr{J})$. Let $d \in D(\mathscr{J}, \mathscr{J})$. Then the map $R(d) : \mathscr{L}(\mathscr{J}) \to \mathscr{L}(\mathscr{J})$ given by

$$R_a R(d) = R_{ad}, \quad [R_a, R_b]\, R(d) = [R_{ad}, R_b] + [R_a, R_{bd}]$$

is a derivation. Define $K_s(d)$: $K_s(\mathscr{J}) \to K_s(\mathscr{J})$ by

$$aK_s(d) = ad, \qquad a'K_s(d) = (ad)', \qquad lK_s(d) = lR(d)$$

where

$$a \in \mathscr{J}, \qquad a' \in \mathscr{J}', \qquad l \in \mathscr{L}(\mathscr{J}).$$

LEMMA 1. *$K_s(d) \in D(K_s(\mathscr{J}), K_s(\mathscr{J}))$ and commutes with π.*

Proof.

$$\begin{aligned}
(a\Delta b)\, K_s(d) &= R_{a,b} R(d) - [R_a, R_b]\, R(d) \\
&= R_{(a,b)d} - [R_{ad}, R_b] - [R_a, R_{bd}] \\
&= R_{ad,b} + R_{a,bd} - [R_{ad}, R_b] - [R_a, R_{bd}] \\
&= ad\Delta b + a\Delta(bd) \\
&= (aK_s(d))\, \Delta b + a\Delta(bK_s(d)).
\end{aligned}$$

$$\begin{aligned}
[a, l]\, K_s(d) &= (al)\, d = adl + a(lR(d)) \\
&= [aK_s(d), l] + [a, lK_s(d)].
\end{aligned}$$

These identities are sufficient to prove $K_s(d)$ a derivation, and commutativity with involution is obvious from the definitions. ∎

Schafer [23] has shown that if $\mathscr{J}$ is Jordan with 1 over a field of characteristic $\neq 2$, then a derivation $d \in D(\mathscr{J}, \mathscr{J})$ is in $\mathscr{L}(\mathscr{J})$ if and only if

$$d = \sum_i [a_i, \ , b_i],$$

where $[a_i, , b_i]$ denotes the associator of a_i and b_i. Hence, if M is a unital bimodule for such an algebra, $d \in D(\mathscr{J}, M)$ is in $\mathscr{L}(E(\mathscr{J}, M))$ if and only if

$$d = \sum_i [a_i, , m_i] = \sum_i [R_{a_i}, R_{m_i}], \qquad a_i \in \mathscr{J}, \qquad m_i \in M.$$

The classical inner derivation functor I is defined by:

$$I(\mathscr{J}, M) = D(\mathscr{J}, M) \cap \mathscr{L}(E(\mathscr{J}, M)).$$

Its properties are discussed in Glassman [26].

LEMMA 2. *Let $d \in D(\mathscr{J}, \mathscr{J})$. Then $K_s(d)$ is inner if and only if $d \in [R(\mathscr{J}), R(\mathscr{J})]$.*

Proof Suppose $xK_s(d) = [b, x]$ for all $x \in K_s(\mathscr{J})$ where $b = a_0 + b_0' + l_0$. Pick $x = a + b' + l$. Then, from the manner in which derivations were lifted from $\mathscr{J}$ to $K_s(\mathscr{J})$, ad is the first component of $xK_s(d)$; that is, $ad = a_0 l - a l_0$. But since x is arbitrary, this must hold for all $l \in \mathscr{L}(\mathscr{J})$, i.e., for all $l_1, l_2, a_0 l_1 = a_0 l_2$. Since characteristic $\mathscr{J} \neq 2$, we may choose $l_1 = R_1$, $l_2 = 2R_1$ and obtain $a_0 = 2a_0$. Hence $a_0 = 0$. Similarly $b_0 = 0$; so $b = l_0$.

Since $1K_s(d) = 0$, $l_0 \neq R_a$ for any $a \neq 0$, $a \in \mathscr{J}$, nor does $l_0 = R_a + [R_b, R_c]$ for $a \neq 0$ in $\mathscr{J}$. Hence $l_0 \in [R(\mathscr{J}), R(\mathscr{J})]$ and $K_s(d) = l_0$. The converse is clear. ∎

Let M be a bimodule for $\mathscr{J}$, $d \in D(\mathscr{J}, M)$. Then d can be extended to $d \in D(E(\mathscr{J}, M), E(\mathscr{J}, M))$ by defining $d/M = 0$, and then to

$$K_s(d) \in D(K_s(E(\mathscr{J}, M)), K_s(E(\mathscr{J}, M))).$$

A check of definitions reveals that $K_s(d)/K_s(\mathscr{J}) \in D(K_s(\mathscr{J}), i_s(M))$ and commutes with π. We write $K_s(d)$ for this restriction. As an immediate corollary to the previous lemma, we have

COROLLARY 1. *Let $d \in D(\mathscr{J}, M)$. Then $K_s(d)$ is inner if and only if $d \in [R(\mathscr{J}), R(M)]$.*

By this corollary the map

$$d \to K_s(d): \frac{D(\mathscr{J}, M)}{I(\mathscr{J}, M)} \to \frac{D(K_s(\mathscr{J}), i_s(M))}{I(K_s(\mathscr{J}), i_s(M))}$$

is a monomorphism. We have seen that the image of this map is contained in

$$\frac{D((K_s(\mathscr{J}), \pi), (i_s(M), \pi))}{I((K_s(\mathscr{J}), \pi), (i_s(M), \pi))}.$$

We want to show that the map above is also a epimorphism. For this, we show that if $D \in D(K_s(\mathscr{J}), i_s(M))$, there is a $d \in I(K_s(\mathscr{J}), i_s(M))$ such that $D - d$ is the image of an element of $D(\mathscr{J}, M)$; and, further, that if D commutes with π, then d may be chosen to commute with π.

LEMMA 3. *If* $D_1, D_2 \in D(K_s(\mathscr{J}), i_s(M))$, $\mathscr{J}D_k \subseteq M$, $\mathscr{J}'D_k \subseteq M'$, $k = 1,2$, $D_1/\mathscr{J} = D_2/\mathscr{J}$, *then* $D_1 = D_2$.

Proof. We prove the equivalent statement that if $D \in D(K_s(\mathscr{J}), i_s(M))$, $\mathscr{J}D \subseteq M$, $\mathscr{J}'D \subseteq M'$, then D is determined by $D/\mathscr{J}$.

Since $[\mathscr{J}, \mathscr{J}] = \mathscr{L}(\mathscr{J})$, we have $\mathscr{L}(\mathscr{J})\, D \subseteq \mathscr{L}(\mathscr{J})$. In the relation

$$[a, R_b]\, D = [aD, R_b] + [a, R_bD],$$

both $[a, R_b]\, D$ and $[aD, R_b]$ are assumed known. Hence the action of $R_bD \in \mathscr{L}(\mathscr{J})$ on $\mathscr{J}$ is determined, and so R_bD is determined for all $b \in \mathscr{J}$. This, in turn, determines $[R_a, R_b]\, D$ for all a, b in $\mathscr{J}$. Now, D is determined on $\mathscr{L}(\mathscr{J})$.

Again,

$$[1, b']\, D = [1D, b'] + [1, b'D],$$

and since $[1, b'] \in \mathscr{L}(\mathscr{J})$, $[1, b']\, D$ and $[1D, b']$ are known. This determines $[1, b'D]$, and, since $b'D \in \mathscr{J}$, $[1, b'D] = R_{b'D}$. But $R_{b'D}$ determines $b'D$. ∎

The equation

$$[1, R_b]\, D = [1D, R_b] + [1, R_bD]$$

implies that if $1D = 0$ then $bD = [1, R_bD]$. But $bD = [1, R_{bD}]$ so, in this case, $R_bD = R_{bD}$.

LEMMA 4. *If* $D \in D(K_s(\mathscr{J}), i_s(M))$, $1D = 1'D = 0$, *then* $D = K_s(d)$ *for some* $d \in D(\mathscr{J}, M)$.

Proof.

$$R_1D = [1, 1']\, D = [1D, 1'] + [1, 1'D] = 0.$$

For $a \in \mathscr{J}$,

$$ad = [a, R_1]\, D = [aD, R_1].$$

Then, since $[i_s(M), R_1] \subseteq M \oplus M'$, $\mathscr{J}D \subseteq M \oplus M'$. But $[aD, 1] = [a, 1]D = 0$ because $[a, 1] = 0$. This implies $aD \subseteq M$ and then $\mathscr{J}D \subseteq M$, since $[m', 1] = -R_m \neq 0$ for all $m' \in M'$. Similarly $\mathscr{J}'D \subseteq M'$.

Define $d : \mathcal{J} \to M$, a linear map, by $ad = aD$ for all $a \in \mathcal{J}$. Then

$$\begin{aligned}(a.b)\, d &= (aR_b)\, d = [a, R_b]\, D = [aD, R_b] + [a, R_bD] \\ &= [aD, R_b] + [a, R_{bD}] \quad \text{(by the remark following the previous lemma)} \\ &= ad.b + a.(bd).\end{aligned}$$

Therefore $d \in D(\mathcal{J}, M)$.

Since $K_s(d)/\mathcal{J} = D/\mathcal{J}$ by definition, the previous lemma shows that $K_s(d) = D$. ∎

LEMMA 5. *Let $D \in D(K_s(\mathcal{J}), i_s(M))$. Then there is a $d \in I(K_s(\mathcal{J}), i_s(M))$ such that*

$$1(D - d) = 1'(D - d) = 0$$

If D commutes with π, then d may be taken to commute with π.

Proof. Write

$$1D = m_1 + m_2' + R_{m_3} + \sum [R_{a_4}, R_{m_4}].$$

Then

$$1D = [1, R_1]\, D = [1D, R_1] + [1, R_1D].$$

Expanding both sides:

$$m_1 + m_2' + R_{m_3} + \sum [R_{a_4}, R_{m_4}] = m_1 - m_2' + [1, R_1D].$$

Hence

$$[1, R_1D] = 2m_2' + R_{m_3} + \sum [R_{a_4}, R_{m_4}].$$

But

$$[1, i_s(M)] \subseteq M + R(M).$$

Since we are working over a field of characteristic $\neq 2$, this implies

$$m_2 = 0 \qquad \text{and} \qquad \sum [R_{a_4}, R_{m_4}] = 0;$$

then

$$1D = m_1 + R_{m_3}, \quad [1, R_1D] = R_{m_3}.$$

The last implies

$$R_1D = n_1 + m_3' + \sum [R_{b_4}, R_{n_4}]$$

for some $n_1 \in M$, $\{n_4\} \subseteq M$, $\{b_4\} \subseteq \mathcal{J}$.

Now

$$R_1D = [1, 1']\, D = [1D, 1'] + [1, 1'D],$$

and expanding:

$$n_1 + m_3' + \sum [R_{b_4}, R_{n_4}] = R_{m_1} + m_3' + [1, 1'D].$$

Hence

$$[1, 1'D] = n_1 - R_{m_1} + \sum [R_{b_4}, R_{n_4}].$$

Again, since

$$[1, i_s(M)] \subseteq M \oplus R(M),$$

this implies

$$\sum [R_{b_4}, R_{n_4}] = 0$$

and

$$1'D = R_{n_1} - m_1' + k_1 + \sum [R_{c_4}, R_{k_4}]$$

for some $k_1 \in M$, $\{k_4\} \subseteq M$, $\{c_4\} \subseteq \mathscr{J}$.

So far we have

$$1D = m_1 + R_{m_3}$$

$$1'D = k_1 - m_1' + R_{n_1} + \sum [R_{c_4}, R_{k_4}]$$

$$R_1 D = n_1 + m_3'.$$

$1'D = [R_1, 1']\, D = [R_1 D, 1'] + [R_1, 1'D]$ and expanding:

$$k_1 - m_1' + R_{n_1} + \sum [R_{c_4}, R_{k_4}] = [R_1 D, 1'] - k_1 - m_1'.$$

Since

$$[i_s(M), 1'] \subseteq M' + R(M)$$

and the characteristic is not 2, this gives $k_1 = 0$ and $\Sigma[R_{c_4}, R_{k_4}] = 0$. Then

$$1'D = R_{n_1} - m_1'.$$

Consider the inner derivation

$$(R_{m_1} + m_3' - n_1)_R.$$

$$1(R_{m_1} + m_3' - n_1)_R = m_1 + R_{m_3} = 1D;$$

and

$$1'(R_{m_1} + m_3' - n_1)_R = -m_1' + R_{n_1} = 1'D.$$

Therefore, this is the required inner derivation.

Finally, if D commutes with π,

$$R_{n_1} - m_1' = 1'D = (1D)' = (m_1 + R_{m_3})' = m_1' - R_{m_3}.$$

Hence $m_1 = 0$, $n_1 = -m_3$, $1D = R_{m_3}$, and the inner derivation $(m_3 + m_3')_R$ clearly commutes with π. ∎

THEOREM 1. *For any module M over a finite-dimensional Jordan algebra $\mathscr{J}$,*

$$\frac{D(\mathscr{J}, M)}{I(\mathscr{J}, M)} \simeq \frac{D(K_s(\mathscr{J}), i_s(M))}{I(K_s(\mathscr{J}), i_s(M))} \simeq \frac{D((K_s(\mathscr{J}), \pi), (i_s(M), \pi))}{I((K_s(\mathscr{J}), \pi), (i_s(M), \pi))}.$$

Proof. As previously noted, we need only show that the monomorphisms already constructed are onto. Let $D \in D(K_s(\mathscr{J}), i_s(M))$. By Lemma 5, there exists $d \in I(K_s(\mathscr{J}), i_s(M))$ such that

$$1(D - d) = 1'(D - d) = 0.$$

By Lemma 4, $D - d$ is in the image of $D(\mathscr{J}, M)$. Finally, we have seen that if D commutes with π, we may choose d to commute with π. ∎

As a corollary to this theorem, we have a new proof of the well known

COROLLARY 2. *If $\mathscr{J}$ is semisimple Jordan of characteristic 0, M a unital bimodule for $\mathscr{J}$, then all derivations from $\mathscr{J}$ to M are inner.*

Proof. $\mathscr{J}$ semisimple implies $K_s(\mathscr{J})$ semisimple. M unital implies $[i_s(m), K_s(\mathscr{J})] \neq 0$. Then from standard results of Lie algebras of characteristic 0 (Jacobson [14]),

$$\frac{D(K_s(\mathscr{J}), i_s(M))}{I(K_s(\mathscr{J}), i_s(M))} = 0.$$

Therefore,

$$\frac{D(\mathscr{J}, M)}{I(\mathscr{J}, M)} = 0. \quad ∎$$

ISOTOPIC JORDAN ALGEBRAS

Let $x, b, y \in \mathscr{J}$.

DEFINITION. $\{xby\} = (x \cdot b) \cdot y + (b \cdot y) \cdot x - (x \cdot y) \cdot b,$

$$U_a = 2R_a^2 - R_{a \cdot a}.$$

DEFINITION. The b homotope of $\mathscr{J}$, written $\mathscr{J}^{(b)}$, is the Jordan algebra on

the vector space $\mathcal{J}$ with multiplication $x._b y = \{xby\}$. If b is invertible in $\mathcal{J}$, then $\mathcal{J}^{(b)}$ is called the *b isotope* of $\mathcal{J}$.

DEFINITION. Two Jordan algebras $\mathcal{J}_1$, $\mathcal{J}_2$ are *homotopic*, if $\mathcal{J}_1 \simeq \mathcal{J}_2^{(b)}$ for some $b \in \mathcal{J}_2$. They are isotopic if b is invertible in $\mathcal{J}_2$.

The final topic of this section is the relation between cohomology modules of isotopic Jordan algebras. Let $\mathcal{J}, M$ be as always, $b \in \mathcal{J}$ an invertible element, $E(\mathcal{J}, M)$ the split null extension with elements written (a, m), $a \in \mathcal{J}$, $m \in M$. We can form $E(\mathcal{J}, M)^{(b)}$, the b isotope. An easy calculation shows that

$$\begin{aligned}(a_1, m_1).{}_b\,(a_2, m_2) &= \{(a_1, m_1)(b, 0)(a_2, m_2)\} \\ &= (\{a_1 b a_2\}, \{m_1 b a_2\} + \{a_1 b m_2\}) \\ &= (a_1 .{}_b a_2, m_1 .{}_b a_2 + a_1 .{}_b m_2).\end{aligned}$$

Thus, there is a one-to-one correspondence (since isotopy is a symmetric relation) between bimodules for $\mathcal{J}$ and those for $\mathcal{J}^{(b)}$. If M is a bimodule for $\mathcal{J}$, the corresponding bimodule $M^{(b)}$ for $\mathcal{J}^{(b)}$ is on the vector space M with multiplication

$$a._b m = m._b a = \{abm\}.$$

Let I be the classical inner derivation functor. For any bimodule M, we want to show that $H_I^n(\mathcal{J}, M) \simeq H_I^n(\mathcal{J}^{(b)}, M^{(b)})$ for all $n \geqslant 1$ where $\{H^n\}$ are the cohomology modules as defined in Glassman [26]. As may have been expected, one tool to be used is the Lie algebra $K_s(\mathcal{J})$ and bimodule $i_s(M)$. The map

$$\varphi : a + c' + l \to a + (cU_b. - 1)' + l$$

is an isomorphism of $K_s(\mathcal{J})$ with $K_s(\mathcal{J}^{(b)})$ and of $i_s(M)$ as a $K_s(\mathcal{J})$-bimodule with $i_s(M^{(b)})$ as a $K_s(\mathcal{J}^{(b)})$-bimodule.

LEMMA 6.

$$\frac{D(\mathcal{J}, M)}{I(\mathcal{J}, M)} \simeq \frac{D(\mathcal{J}^{(b)}, M^{(b)})}{I(\mathcal{J}^{(b)}, M^{(b)})}.$$

Proof. We already know that

$$\frac{D(\mathcal{J}, M)}{I(\mathcal{J}, M)} \simeq \frac{D(K_s(\mathcal{J}), i_s(M))}{I(K_s(\mathcal{J}), i_s(M))}$$

and

$$\frac{D(\mathcal{J}^{(b)}, M^{(b)})}{I(\mathcal{J}^{(b)}, M^{(b)})} \simeq \frac{D(K_s(\mathcal{J}^{(b)}), i_s(M^{(b)}))}{I(K_s(\mathcal{J}^{(b)}), i_s(M^{(b)}))}.$$

It will be enough to show that

$$\frac{D(K_s(\mathscr{J}), i_s(M))}{I(K_s(\mathscr{J}), i_s(M))} \simeq \frac{D(K_s(\mathscr{J}^{(b)}), i_s(M^{(b)}))}{I(K_s(\mathscr{J}^{(b)}), i_s(M^{(b)}))}.$$

If $d \in D(K_s(\mathscr{J}), i_s(M))$, the map $\varphi^{-1} d\varphi \in D(K_s(\mathscr{J}^{(b)}), i_s(M^{(b)}))$ since the composition of an isomorphism with a derivation is a derivation. Since φ is an isomorphism, we have

$$D(K_s(\mathscr{J}), i_s(M)) \simeq D(K_s(\mathscr{J}^{(b)}), i_s(M^{(b)})),$$

and we need only show that this preserves inner derivations.

Consider

$$c \to [c, m], \qquad m \in i_s(M).$$

For $c' \in K_s(\mathscr{J}^{(b)})$,

$$c'\varphi^{-1} d\varphi = [c'\varphi^{-1}, m]\,\varphi = [c', m\varphi] \in I(K_s(\mathscr{J}^{(b)}), i_s(M^{(b)})). \quad \blacksquare$$

LEMMA 7. $H^2(\mathscr{J}, M) \simeq H^2(\mathscr{J}^{(b)}, M^{(b)})$.

Proof. Suppose $0 \to M \to \mathscr{L} \to \mathscr{J} \to 0$ is a short singular extension of M by $\mathscr{J}$. $b \in \mathscr{J}$ is invertible and, by a theorem in Jacobson [15], $(b, 0) \in \mathscr{L}$ is invertible. Suppose multiplication in $\mathscr{L}$ is given by

$$(a_1, m_1) \cdot (a_2, m_2) = (a_1 \cdot a_2, m_1 \cdot a_2 + m_1 \cdot a_2 + f(a_1, a_2)),$$

f a Jordan 2-cocycle. Then, an easy computation shows that multiplication in $\mathscr{L}^{(b,0)}$ is given by

$$(a_1, m_1) \cdot_{(b,0)} (a_2, m_2) = (\{a_1 b a_2\}, \{m_1 b a_2\} + \{a_1 b m_2\} + g(a_1, a_2))$$

where

$$g(a_1, a_2) = f(a_1, b) \cdot a_2 + f(a_1 \cdot b, a_2) + f(b, a_2) \cdot a_1 + f(b \cdot a_2, a_1) - f(a_1, a_2), b - f(a_1, a_2, b).$$

It is clear that the map which associates to the extension $0 \to M \to \mathscr{L} \to \mathscr{J} \to 0$, the extension $0 \to M^{(b)} \to \mathscr{L}^{(b,0)} \to \mathscr{J}^{(b)} \to 0$ is well-defined and respect equivalence. Hence we have a homomorphism

$$\Phi : H^2(\mathscr{J}, M) \to H^2(\mathscr{J}^{(b)}, M^{(b)}).$$

To show that Φ is an isomorphism, we construct an inverse Ψ. Let $(b, 0)^{\cdot -2} = (a, m)$. Then $a = b^{\cdot -2}$. Given an exact sequence $0 \to M^{(b)} \to \mathscr{L}' \to \mathscr{J}^{(b)} \to 0$, we associate the sequence $0 \to M \to \mathscr{L}'^{(a,m)} \to \mathscr{J} \to 0$. Since $\mathscr{J}^{(b)}(b^{\cdot -2}) = \mathscr{J}$, $M^{(b)}(b^{b^{\cdot -2}}) = M$, we can prove as before that this map preserves equivalence and induces $\Phi^{-1} = \Psi$. $\blacksquare$

LEMMA 8. $H^n(\mathscr{J}, M) \simeq H^n(\mathscr{J}^{(b)}, M^{(b)})$ *for all* $n \geqslant 2$.

Proof. It is clear that exact sequences of bimodules for $\mathscr{J}$ and $\mathscr{J}^{(b)}$ coincide. Further, if

$$f \in \mathrm{Hom}_{U(\mathscr{J})}(M_1, M_2),$$

then

$$f \in \mathrm{Hom}_{U(\mathscr{J}^{(b)})}(M_1^{(b)}, M_1^{(b)})$$

without change. Consequently, equivalence of sequences of bimodules is defined independent of whether the bimodules are regarded as being operated upon by $\mathscr{J}$ or by $\mathscr{J}^{(b)}$. When we combine this with the equivalence above for $n = 2$, we have the lemma. ∎

COHOMOLOGY OF JORDAN MATRIX ALGEBRAS

We are now ready to begin the program outlined in the introduction.

Let $(\mathscr{A}, r)$ be an associative algebra with unit and involution over a field K of characteristic $\neq 2$, $n \geqslant 4$. The involution $A_n \to (A_n{}^r)^t$, where $A_n{}^r$ is the matrix formed by applying r to each component of A_n and t denotes transpose, is called the standard involution. If $\{a_i\}^n$ are invertible and symmetric in $\mathscr{A}$,

$$A_n \to \mathrm{diag}(a_1, \ldots, a_n)(A_n{}^r)^t \,\mathrm{diag}(a_1^{-1}, \ldots, a_n^{-1})$$

is an involution called canonical. If R is any canonical involution of $\mathscr{A}_n$, $\mathscr{H}_n(\mathscr{A}, R)$, the subalgebra of $\mathscr{A}_n{}^+$ of R-symmetric elements, is Jordan. We write

$$a[ij] = ae_{ij} + a_i(ar)a_j^{-1}e_{ji} \in \mathscr{H}_n(\mathscr{A}, R), \qquad a \in \mathscr{A}.$$

Jacobson [15] has shown that the category of unital Jordan bimodules for $\mathscr{H}_n(\mathscr{A}, R)$ and the category of unital associative bimodules with involution for $(\mathscr{A}, R)$ are equivalent. For the map Γ on objects, let (N, r) be an associative bimodule with involution for $(\mathscr{A}, r)$. Then $(E(\mathscr{A}, N), r)$ is an associative algebra with unit and involution r extending that on $\mathscr{A}$, and we form $\mathscr{H}_n(E(\mathscr{A}, N), R)$, a Jordan algebra with unit. Define

$$\Gamma(N, r) = N_n \cap \mathscr{H}_n(E(\mathscr{A}, N), R),$$

a unital Jordan bimodule for $\mathscr{H}_n(\mathscr{A}, R)$.

Conversely, if M is a unital Jordan bimodule for $\mathscr{H}_n(\mathscr{A}, R)$; $E(\mathscr{H}_n(\mathscr{A}, R), M)$ is a Jordan algebra with unit and, by the coordinatization theorem (Jacobson [15]),

$$E(\mathscr{H}_n(\mathscr{A}, R), M) \simeq \mathscr{H}_n(\mathscr{B}, \bar{R})$$

for some associative $(\mathscr{B}, \bar{r})$. It can be seen that

$$(\mathscr{B}, \bar{r}) \simeq (E(\mathscr{A}', M'), r')$$

where

$$(\mathscr{A}', r') \simeq (\mathscr{A}, r),$$

(M', r') a unital associative bimodule with involution for $(\mathscr{A}', r')$. Then $M = \Gamma(M', r')$.

For any algebra $\mathscr{A}$, $U(\mathscr{A})$ is the universal enveloping algebra of $\mathscr{A}$, and $U_1(\mathscr{A})$ is the unital universal enveloping algebra of $\mathscr{A}$. If $\mathscr{A}$ has involution r, $(U(\mathscr{A}), \bar{\sigma})$ is the universal enveloping algebra of $\mathscr{A}$ with involution $\bar{\sigma}$.

For morphisms, if $(N_1, r_1), (N_2, r_2)$ are unital bimodules for $(\mathscr{A}, r)$, $\eta \in \mathrm{Hom}_{(U^1(\mathscr{A}),\bar{\sigma})}((N_1, r_1), (N_2, r_2))$, then η induces

$$\Gamma(\eta) \in \mathrm{Hom}_{U(\mathscr{H}_n(\mathscr{A},R))}(\Gamma(N_1, r_1), \Gamma(N_2, {}_2))$$

by applying η to each component. Conversely, if

$$\bar{\eta} \in \mathrm{Hom}_{U(\mathscr{H}_n(\mathscr{A},R))}(\Gamma(N_1, r_1), \Gamma(N_2, r_2)),$$

then $\bar{\eta} = \Gamma(\eta)$ for some

$$\eta \in \mathrm{Hom}_{(U_1(\mathscr{A}),\sigma)}((N_1, r_1), (N_2, r_2)).$$

Jacobson also shows the following. Suppose $(\mathscr{A}_1, r_1)$, $(\mathscr{A}_2, r_2)$ are associative with unit and involution; $\mathscr{H}_n(\mathscr{A}_1, R_1)$, $\mathscr{H}_n(\mathscr{A}_2, R_2)$ are the Jordan algebras given by canonical involutions using the matrices of invertible symmetric elements $\mathrm{diag}(a_i)$, $\mathrm{diag}(b_i)$, respectively. Suppose that

$$\eta : (\mathscr{A}_1, r_1) \to (\mathscr{A}_2, r_2)$$

is a morphism af algebras with involution such that $a_i\eta = b_i$ for all i. Then η induces

$$\bar{\eta} : \mathscr{H}_n(\mathscr{A}_1, R_1) \to \mathscr{H}_n(\mathscr{A}_2, R_2),$$

a morphism of algebras, by applying η to each component of the matrix. Conversely, if

$$\bar{\eta} : \mathscr{H}_n(\mathscr{A}_1, R_1) \to \mathscr{H}_n(\mathscr{A}_2, R_2)$$

is a morphism of algebras such that

$$1_{\mathscr{A}_1}[ij]\bar{\eta} = 1_{\mathscr{A}_2}[ij]$$

for all i, j, then $\bar{\eta}$ is induced as above.

Assume, until otherwise specified, that R is standard. Let $d \in D((\mathscr{A}, r), (N, r))$. Then d induces $\delta \in \mathrm{Hom}((E(\mathscr{A}, N), r), (E(\mathscr{A}, N), r))$ of algebras by $(a, m)\,\delta = (a, ad)$, $m \in N$. Since $1\delta = 1$, we have

$$\bar{\delta} \in \mathrm{Hom}(\mathscr{H}_n(E(\mathscr{A}, N), R), \mathscr{H}_n(E(\mathscr{A}, N), R))$$

induced as above. Let $A \in \mathscr{A}_n \cap \mathscr{H}_n(E(\mathscr{A}, N), R)$. Then $A\bar{\delta} = (A, Ad)$, Ad the matrix of elements of N formed by applying d componentwise to A. Since $\bar{\delta}$ is a homomorphism of algebras, this implies that $\Gamma(d)$, the map formed by restricting $\bar{\delta}$ to

$$\mathscr{A}_n \cap \mathscr{H}_n(E(\mathscr{A}, N), R) \simeq \mathscr{H}_n(\mathscr{A}, R)$$

and considering only the second component of the image, is in

$$D(\mathscr{H}_n(\mathscr{A}, R), \Gamma(N, r)).$$

LEMMA 9. *Let $D \in D(\mathscr{H}_n(\mathscr{A}, R), \Gamma(N, r))$. Then there exists a*

$$d \in D((\mathscr{A}, r), (N, r))$$

such that $D = \Gamma(d)$, if and only if $1[ij]\,D = 0$ for all i, j.

Proof. If $D = \Gamma(d)$, the result is clear. Conversely, suppose $1[ij]\,D = 0$. Then $D \in D(\mathscr{H}_n(\mathscr{A}, R), \Gamma(N, r))$ can be extended to

$$D : E(\mathscr{H}_n(\mathscr{A}, r), \Gamma(N, r)) \to E(\mathscr{H}_n(\mathscr{A}, R), \Gamma(N, r)),$$

a homomorphism of algebras, by defining $(a, m)\,D = (a, aD)$, $m \in \Gamma(N, r)$. This extension maps $1[ij]$ to $1[ij]$ and so, by previous considerations, arises from $\delta \in \mathrm{Hom}((E(\mathscr{A}, N), r), (E(\mathscr{A}, N), r))$ with $1\delta = 1$, $a\delta = (a, ad)$ for some $d \in D((\mathscr{A}, r), (N, r))$. Then $D = \Gamma(d)$. ∎

If (N, r) is a unital associative bimodule for $(\mathscr{A}, r)$, then (N_n, R) is a unital bimodule for $(\mathscr{A}_n, R)$ under associative multiplication of matrices. Let $a \in \mathscr{A}_n$ be symmetric, $m \in N_n$ skew. Then

$$(am - ma)R = (mR)(aR) - (aR)(mR) = (-m)a - a(-m) = am - ma;$$

that is, $m_R - m_L$ (associative multiplication) is a derivation of $\mathscr{A}_n$ into N_n which preserves symmetric elements. By restriction to the symmetric elements we get

$$m_R - m_L \in D(\mathscr{H}_n(\mathscr{A}, R), (N, r)),$$

for all m skew in N_n.

Suppose $\Gamma(N_1, r) \xrightarrow{\pi} \Gamma(N_2, r) \to 0$ is exact. We know that $\pi = \Gamma(\eta)$ for some $\eta : (N_1, r) \to (N_2, r)$, and η is clearly an epimorphism. Then η induces an epimorphism $\bar{\eta} : (N_{1_n}, R) \to (N_{2_n}, R)$. Suppose $m_2 \in (N_{2_n}, R)$, m_2 skew, $m_1 \in (N_{1_n}, R)$ with $m_1\bar{\eta} = m_2$. Then $\frac{1}{2}(m_1 - m_1R) \in (N_{1_n}, R)$ is skew, and is carried into m_2 by $\bar{\eta}$.

Thus we see that

$$\{d \in D(\mathscr{H}_n(\mathscr{C}, R), \Gamma(N_1, r)) |$$
$$d = m_{1R} - m_{1L},\ m_1 \text{ skew in } (N_{1_n}, R)\} \to \{d \in D(\mathscr{H}_n(\mathscr{C}, R), \Gamma(N_2, r)) |$$
$$d = m_{2R} - m_{2L},\ m_2 \text{ skew in } (N_{2_n}, R)\} \to 0$$

is exact.

If we define, for any $\mathscr{H}_n(\mathscr{C}, R)$-bimodule $\Gamma(N, r)$, $J(\mathscr{H}_n(\mathscr{C}, R), \Gamma(N, r))$ to be the K vector space of derivations of the form $n_R - n_L$, n skew in (N_n, R), then by the remarks above, J is an inner derivation functor.

Our aim is to prove that

$$H_I^1((\mathscr{C}, r), (N, r)) \simeq H_J^1(\mathscr{H}_n(\mathscr{C}, R), \Gamma(N, r)).$$

To that end, the next several lemmas will show that if

$$D \in D(\mathscr{H}_n(\mathscr{C}, R), \Gamma(N, r)),$$

then there is a

$$\bar{D} \in J(\mathscr{H}_n(\mathscr{C}, R), \Gamma(N, r))$$

such that

$$1[ij](D - \bar{D}) = 0$$

for all i, j, and therefore that

$$D - \bar{D} = \Gamma(d)$$

for some

$$d \in D((\mathscr{C}, r), (N, r)).$$

By an easy computation, any classical inner derivation

$$\Sigma[R_{a_i}, R_{m_i}]$$

can be written

$$\Sigma[a_i, m_i]_R - \Sigma[a_i, m_i]_L,$$

as a Lie triple product. Here, the products am, ma are not the Jordan elements $a \cdot m$, $m \cdot a$, but are formed in the associative bimodule N_n for $\mathscr{C}_n$. Since a_i, m_i are skew in $\mathscr{C}_n$, N_n, respectively, $[a_i, m_i]$ is also skew. Hence

$$I(\mathscr{H}_n(\mathscr{C}, R), \Gamma(N, r)) \subseteq J(\mathscr{H}_n(\mathscr{C}, R), \Gamma(N, r)).$$

We reproduce the multiplication table for Jordan matrix algebras with standard involvution.

$$2a[ij] \cdot b[jk] = ab[ik] \qquad i, j, k \text{ distinct}$$
$$2a[ii] \cdot b[ij] = (ab + (ar)\, b)[ij] \qquad i \neq j$$

$$2a[ij] \cdot b[ji] = ab[ii] + [ba]jj \qquad i \neq j$$
$$a[ii] \cdot b[ii] = (a + ar) \cdot (b + br)[ii]$$
$$a[ij] \cdot b[kl] = 0 \qquad \text{neither } i \text{ nor } j \text{ equal } k \text{ or } l$$
$$a[ij] = (ar)[ji].$$

LEMMA 10. *If* $1[1i]\,D = 0$ *for all* i, *then* $1[ij]\,D = 0$ *for all* i, j.

Proof.

$$i \neq j : 1[ij] = 2(1[1i] \cdot 1[1j]),$$

so

$$1[ij]D = 2(1[1i]\,D \cdot 1[1j]) + 2(1[1i] \cdot 1[1j]\,D) = 0.$$
$$i = j \neq 1 : 1[jj] = 2(1[1j] \cdot 1[1j]) - 1[11].$$

As before

$$(1[1j] \cdot 1[1j])\,D = 0.$$

$1[11]\,D = 0$ by assumption. Therefore $1[jj]\,D = 0$. ∎

We write

$$N[ij] = \{n[ij] \in \Gamma(N, r) \mid n \in N\}$$
$$= \{ne_{ij} + (nr)e_{ji} \mid n \in N\}$$

LEMMA 11. $1[11]\,D \in \sum_{j \neq 1} N[1j]$.

Proof. The equation

$$\tfrac{1}{2}[11] \cdot \tfrac{1}{2}[11] = \tfrac{1}{2}[11]$$

yields

$$\tfrac{1}{2}[11] \cdot (1[11]\,D) = \tfrac{1}{2}[11]\,D.$$

This says that $1[11]\,D$ lies in the half-space of the idempotent $\frac{1}{2}[11]$. Since $1[11]\,D \in \Gamma(N, r)$ we have

$$1[11]D \in \sum_{j \neq 1} N[1j]. \quad ∎$$

LEMMA 12. *If* $1[11]\,D = 0, j \neq 1$, *then* $1[1j]\,D \in \sum_{i \neq 1} N[1i]$.

Proof. The equation $1[1j] = 1[11] \cdot 1[1j]$ yields

$$1[1j]\,D = 1[11]\,D \cdot 1[1j] + 1[11] \cdot (1[1j]\,D).$$

Rewriting:

$$\tfrac{1}{2}[1j]\,D = \tfrac{1}{2}[11] \cdot (1[1j]\,D),$$

that is, $1[1j]\,D$ lies in the half-space of the idempotent $\frac{1}{2}[11]$. Therefore

$$1[1i]D \in \sum_{i \neq 1} N[1i]. \quad \blacksquare$$

LEMMA 13. *Let* $D \in D(\mathscr{H}_n(\mathscr{O}, R), \Gamma(N, r))$ *be such that* $1[1j]\,D$ *has* $[1i]$ *component* 0 *for all* $j \geqslant 1, i > 1$. *Then* $1[ij]\,D = 0$ *for all* i, j.

Proof. By Lemma 11,

$$1[11]D \in \sum_{i \neq 1} N[1i],$$

implying $1[11]\,D = 0$. With this, by Lemma 12,

$$1[1j]D \in \sum_{i \neq 1} N[1i] \qquad \text{for all} \quad j > 1,$$

implying $1[1j]\,D = 0$. Then, by Lemma 10,

$$1[ij]\,D = 0 \qquad \text{for all} \quad i, j. \quad \blacksquare$$

LEMMA 14. *For* $i \neq 1$, *the coefficient of* $[1i]$ *in* $1[1i]\,D$ *is a skew element of* (N, r).

Proof. By applying Lemma 11 to the idempotents $\frac{1}{2}[11]$, $\frac{1}{2}[ii]$, we see that $(1[1i]^{\cdot 2})\,D = (\frac{1}{2}[11] + \frac{1}{2}[ii])\,D$ has $[11]$ component 0. But this is just $2[1i] \cdot (1[1i]\,D)$ and the coefficient of $[11]$ of this product is exactly the coefficient of $[1i]$ in $1[1i]\,D$, say m. Then $m[11] = 0$. Expanding:

$$(m + mr)\, e_{11} = 0,$$

implying $m = -mr$. $\blacksquare$

THEOREM 2. *Let* $D \in D(\mathscr{H}_n(\mathscr{O}, R), \Gamma(N, r))$. *Then there is a*

$$\bar{D} \in J(\mathscr{H}_n(\mathscr{O}, R), \Gamma(N, r))$$

such that

$$1[ij](D - \bar{D}) = 0$$

for all i, j. *Hence there is a* $d \in D((\mathscr{O}, r), (N, r))$ *such that*

$$D - \bar{D} = \Gamma(d).$$

Proof. By Lemma 13, it will be sufficient to find $\bar{D}$ such that the $[1i]$ component of $1[11](D - \bar{D})$ is 0 for all $i > 0$ and the $[1i]$ component of $1[1j](D - \bar{D})$ is 0 for all $i > 1, j > 1$.

Write

$$1[11]D = \sum_{k=2}^{n} m_k[1k]$$

and let

$$D_1 = \sum_{k=2}^{n} [2[1j],\ ,\ m_k[jk]],$$

the associator; where, for each $k, j \neq 1, k$. D_1 is independent of the choices of $j \neq 1, k$. We see that

$$1[11]D_1 = \sum_{k=2}^{n} m_k[1k]$$

and

$$D_1 \in I(\mathscr{H}_n(\mathscr{O}, R), \Gamma(N, r)) \subseteq J(\mathscr{H}_n(\mathscr{O}, R), \Gamma(N, r)).$$

We now have

$$1[11](D - D_1) = 0$$

and, by the remarks at the beginning of this theorem and Lemmas 12 and 14, we may assume

$$1[1j](D - D_1) = \sum_{k=2}^{n} m_{j,k}[1k]$$

where $m_{j,j}$ is skew. For $j \neq 1$ let

$$D_j = \sum_{k \neq 1,j} [-2[kk],\ ,\ m_{j,k}[jk]].$$

Then

$$1[1j]D_j = \sum_{k \neq 1,j} m_{j,k}[1k], \quad 1[1l]D_j = 0$$

for $l \neq j$. Thus

$$\sum_{l=1}^{n} D_l \in I(\mathscr{H}_n(\mathscr{O}, R), \Gamma(N, r)),\ 1[11]\left(D - \sum_{1}^{n} D_l\right) = 0,$$

and

$$1[1j]\left(D - \sum_{1}^{n} D_l\right) = m_{j,j}[1j],$$

$m_{j,j}$ skew in (N, r).

Consider the matrix $\operatorname{diag}(m_{j,j})$ where $m_{1,1} = 0$. This is skew in (N_n, R) since it has only diagonal entries each of which is skew. Further

$$1[1j](\operatorname{diag}(m_{j,j})) - (\operatorname{diag}(m_{j,j}))\,1[1j] = m_{j,j}[1j]$$

for all $j \neq 1$. Finally

$$D_{n+1} = \operatorname{diag}(m_{j,j})_R - \operatorname{diag}(m_{j,j})_L \in J(\mathscr{H}_n(\mathscr{O}, R), \Gamma(N, r)).$$

Thus, the derivation whose existence is asserted in the theorem is

$$\sum_1^{n+1} D_l\,. \quad \blacksquare$$

It is interesting (but not necessary for the sequel) to examine, here, the case characteristic $K \nmid n$. With D as in the theorem, we consider the restriction of D to $\mathscr{H}_n(K, R)$, the subalgebra of $\mathscr{H}_n(\mathscr{O}, R)$ generated by $\{k[ij] \mid k \in K \text{ all } i, j\}$. $\mathscr{H}_n(K, R)$ is separable and, by a theorem of Harris [7], in case characteristic $K \nmid n$, all derivations of $\mathscr{H}_n(K, R)$ into any bimodule are classically inner; in particular, the restriction of D is such. Hence, we immediately obtain a

$$\bar{D} \in I(\mathscr{H}_n(\mathscr{O}, R), \Gamma(N, r))$$

such that

$$1[ij](D - \bar{D}) = 0$$

for all i, j.

With the analysis of the theorem, however, we can explicitly construct $\bar{D}$. Since the constructions of $D_1, \ldots, D_n$ all yielded derivations in $I(\mathscr{H}_n(\mathscr{O}, R, \Gamma(N, r))$, we are reduced to the case

$$1[1j]\,D = m_{j,j}[1j],\ m_{j,j}$$

skew in (N, r), $1[11]\,D = 0$. Write

$$m_{j,j} = m_j - m_j r,\ m_j \in N$$

and let, for $j \neq 1$,

$$s_j = \left(\frac{-1}{n}\right)(m_2 + m_3 + \cdots + m_{j-1} + m_{j+1} + \cdots + m_n) + \left(\frac{n-1}{n}\right) m_j\,.$$

Define

$$D_{n+1} = \sum_{j \neq 1} [4[1j],\ ,\ s_j[1j]].$$

It is now a tedious but not difficult calculation to verify that D_{n+1} has the required effect.

Recall that all of the preceding was done for R standard. We maintain that assumption.

COROLLARY 3. $H_I^1((\mathcal{A}, r), (N, r)) \simeq H_J^1(\mathcal{H}_n(\mathcal{A}, R), \Gamma(N, r))$.

Proof. By Theorem 2,

$$\Gamma : D((\mathcal{A}, r), (N, r)) \to \frac{D(\mathcal{H}_n(\mathcal{A}, R), \Gamma(N, r))}{J(\mathcal{H}_n(\mathcal{A}, R), \Gamma(N, r))}$$

is onto.

$$I((\mathcal{A}, r), (N, r)) = [m_R - m_L \mid m \text{ skew in } N],$$

and if $d = m_R - m_L$,

$$\Gamma(d) = \operatorname{diag}(m)_R - \operatorname{diag}(m)_L \in J(\mathcal{H}_n(\mathcal{A}, R), \Gamma(N, r)).$$

Hence Γ induces an epimorphism

$$\Gamma : H_I^1((\mathcal{A}, r), (N, r)) = \frac{D((\mathcal{A}, r), (N, r))}{I((\mathcal{A}, r), (N, r))} \to H_J^1(\mathcal{H}_n(\mathcal{A}, R), \Gamma(N, r)).$$

We show that this induced map is one-to-one by showing that if

$$D \in J(\mathcal{H}_n(\mathcal{A}, R), \Gamma(N, r)), \qquad D = \Gamma(d),$$

then $d \in I((\mathcal{A}, r), (N, r))$. From remarks at the beginning of this section, and by definition of J, if

$$D \in J(\mathcal{H}_n(\mathcal{A}, R), \Gamma(N, r)),$$

then $D = B_R - B_L$, $B \in N_n$, where this is associative matrix multiplication. Further, since $D = \Gamma(d)$ for some d, $1[ij]\, D = 0$ for all i, j, and $a[ij]\, D \in N[ij]$ for all $a \in \mathcal{A}$.

Suppose

$$B = \sum_{i,j} m_{ij} e_{ij}, \; m_{ij} \in N.$$

Then for all i,

$$1[ii]\, D = e_{ii}B - Be_{ii} = 0.$$

In particular, the e_{ik} coefficient of $1[ii]\, D$, $i \neq k$, is 0. But this is m_{ik}. Hence $m_{ik} = 0$ for $i \neq k$ and $B = \operatorname{diag}(m_1, \ldots, m_k)$.

Next, $1[ij]\, D = 0$ for all $i \neq j$. That is

$$(e_{ij} + e_{ji})\, B - B(e_{ij} + e_{ji}) = 0,$$

or

$$m_j e_{ij} + m_i e_{ji} - m_i e_{ij} - m_j e_{ji} = 0.$$

Combining:

$$(m_j - m_i)\, e_{ij} + (m_i - m_j)\, e_{ji} = 0.$$

This implies $m_i = m_j = m$ for all i, j; and $D = \Gamma(d)$ where, for $a \in \mathcal{A}$, $ad = am - ma$. Since d is a derivation, m is skew. ∎

LEMMA 15. *Let $\bar{R}$ be any canonical involution. (N, r) is an injective $(\mathscr{A}, r)$ bimodule if and only if*

$$\Gamma(N, r) = N_n \cap \mathscr{H}_n(E(\mathscr{A}, N), R)$$

is an injective $\mathscr{H}_n(\mathscr{A}, \bar{R})$-bimodule.

Proof. Suppose $\Gamma(N, r)$ is injective. We show (N, r) is injective, that is, injective in the category of bimodules with involution. Suppose given the diagram

$$\begin{array}{ccccc} 0 & \longrightarrow & (N_1, r_1) & \longrightarrow & (N_2, r_2) \\ & & \downarrow & & \\ & & (N, r) & & \end{array}$$

of $(\mathscr{A}, r)$ bimodules. This induces a diagram

$$\begin{array}{ccccc} 0 & \longrightarrow & \Gamma(N_1, r_1) & \longrightarrow & \Gamma(N_2, r_2) \\ & & \downarrow & \swarrow_{f} & \\ & & \Gamma(N, r) & & \end{array}$$

of $\mathscr{H}_n(\mathscr{A}, \bar{R})$-bimodules, where f exists since $\Gamma(N, r)$ is injective. But we have seen that any such homomorphism has the form

$$\Gamma(g), g \in \mathrm{Hom}_{(U^1(\mathscr{A}), \bar{\sigma})}((N_2, r_2), (N, r)).$$

Clearly, g completes the original diagram.

The converse is similar. ∎

Now, by Corollary 3, for (N, r) injective, R standard

$$H_J^1(\mathscr{H}_n(\mathscr{A}, r), \Gamma(N, r)) \simeq H_I^1((\mathscr{A}, r), (N, r)) = 0.$$

By Lemma 15, since Γ is an equivalence,

COROLLARY 4. *If R is standard, M an injective bimodule for $\mathscr{H}_n(\mathscr{A}, R)$, then*

$$H_J^1(\mathscr{H}_n(\mathscr{A}, R), M) = 0. \quad ∎$$

We are now ready to extend these results to matrix algebras with canonical involution. Recall that if $\mathscr{J}$ is a Jordan algebra over a field K of characteristic $\neq 2$, M a $\mathscr{J}$-bimodule, we constructed a Lie algebra $K_s(\mathscr{J})$ and a Lie bimodule $i_s(M)$ for it with the property that

$$\frac{D(\mathscr{J}, M)}{I(\mathscr{J}, M)} \overset{\varphi}{\simeq} \frac{D(K_s(\mathscr{J}), i_s(M))}{I(K_s(\mathscr{J}), i_s(M))}$$

where φ is induced by the injection

$$D(\mathscr{J}, M) \to D(K_s(\mathscr{J}), i_s(M)).$$

Further, if $\mathscr{J}^{(b)}$ is isotopic to $\mathscr{J}$, then

$$K_s(\mathscr{J}) \overset{\omega}{\simeq} K_s(\mathscr{J}^{(b)})$$

(but ω is not a map of algebras with automorphism), and there is a bimodule $M^{(b)}$ for $\mathscr{J}^{(b)}$ such that

$$D(K_s(\mathscr{J}), i_s(M)) \overset{\Psi^{-1}}{\simeq} D(K_s(\mathscr{J}^{(b)}), i_s(M^{(b)}))$$

and

$$H_I^1(K_s(\mathscr{J}), i_s(M)) \overset{\Psi^{-1}}{\simeq} H_I^1(K_s(\mathscr{J}^{(b)}), i_s(M^{(b)})).$$

Let R be standard, $\bar{R}$ canonical. We know that

$$\mathscr{H}_n(\mathscr{C}, \bar{R}) \simeq \mathscr{H}_n(\mathscr{C}, R)^{(b)}$$

for some b, and we use the latter form. We write

$$\mathscr{J} = \mathscr{H}_n(\mathscr{C}, R), \qquad \mathscr{J}^{(b)} \cong \mathscr{H}_n(\mathscr{C}, \bar{R}).$$

Let M be a $\mathscr{J}$ bimodule.

Define

$$J(K_s(\mathscr{J}), i_s(M)) \subseteq D(K_s(\mathscr{J}), i_s(M))$$

to be

$$\text{image } (J(\mathscr{J}, M) \to D(K_s(\mathscr{J}), i_s(M))) + I(K_s(\mathscr{J}), i_s(M)).$$

Then

$$J(K_s(\mathscr{J}), i_s(M)) \supseteq I(K_s(\mathscr{J}), i_s(M)).$$

Consider the map

$$\Phi : D(\mathscr{J}^{(b)}, M^{(b)}) \to \frac{D(K_s(\mathscr{J}), i_s(M))}{J(K_s(\mathscr{J}), i_s(M))}$$

given by the composite

$$D(\mathscr{J}^{(b)}, M^{(b)}) \longrightarrow D(K_s(\mathscr{J}^{(b)}), i_s(M^{(b)})) \xrightarrow{\Psi} D(K_s(\mathscr{J}), i_s(M))$$

$$\xrightarrow{\text{proj}} \frac{D(K_s(\mathscr{J}), i_s(M))}{J(K_s(\mathscr{J}), i_s(M))}.$$

Φ is an epimorphism, because it can also be considered as the composite

$$D(\mathscr{J}^{(b)}, M^{(b)}) \xrightarrow{\text{onto}} \frac{D(K_s(\mathscr{J}^{(b)}), i_s(M^{(b)}))}{I(K_s(\mathscr{J}^{(b)}), i_s(M^{(b)}))} \xrightarrow{\simeq}$$

$$\frac{D(K_s(\mathscr{J}), i_s(M))}{I(K_s(\mathscr{J}), i_s(M))} \xrightarrow{\text{onto}} \frac{D(K_s(\mathscr{J}), i_s(M))}{J(K_s(\mathscr{J}), i_s(M))}.$$

DEFINITION. $J(\mathscr{J}^{(b)}, M^{(b)}) = \ker \Phi$.

Note first that all $\mathscr{J}^{(b)}$ bimodules are of the form $M^{(b)}$ for some $\mathscr{J}$ bimodule M, so $J(\mathscr{J}^{(b)}, \ )$ is well-defined. Note also that if $b = 1$, i.e. $R = \bar{R}$, then this definition of $J(\mathscr{J}, \ )$ agrees with the former one. We want to verify, in the general case, that $J(\mathscr{J}^{(b)}, \ )$ is an inner derivation functor; the only part of this that is not obvious is that $J(\mathscr{J}^{(b)}, \ )$ preserves epimorphisms.

With $\mathscr{J}$ as above, let $M_1 \xrightarrow{\pi} M_2 \to 0$ be an exact sequence of $\mathscr{J}$ bimodules. Using the argument about lifting of skew elements that we previously used, it is easy to see that

$$J(K_s(\mathscr{J}), i_s(M_1)) \to J(K_s(\mathscr{J}), i_s(M_2)) \to 0$$

is exact. We have the injection

$$\varphi : D(\mathscr{J}^{(b)}, M_i^{(b)}) \to D(K_s(\mathscr{J}^{(b)}), i_s(M_i^{(b)})).$$

By definition, if

$$d \in D(\mathscr{J}^{(b)}, M_2^{(b)}),$$

then

$$d \in J(\mathscr{J}^{(b)}, M_2^{(b)})$$

if and only if

$$d\varphi\Psi \in J(K_s(\mathscr{J}), i_s(M)).$$

Then there exists

$$\bar{d} \in J(K_s(\mathscr{J}), i_s(M_1))$$

such that $\bar{d} \circ \pi = d\varphi\Psi$.

Letting

$$d' = \bar{d}\Psi^{-1} \in D(K_s(\mathscr{J}^{(b)}, i_s(M_1^{(b)})),$$

we have $d'\Psi \circ \pi = d\varphi\Psi$.

Consider the commutative diagram

$$\begin{array}{ccc}
D(\mathscr{J}^{(b)}, M_1^{(b)}) & \xrightarrow{D(\mathscr{J}^{(b)}, \pi)} & D(\mathscr{J}^{(b)}, M_2^{(b)}) \\
\downarrow \varphi & & \downarrow \varphi \\
D(K_s(\mathscr{J}^{(b)}), i_s(M_1^{(b)})) & \xrightarrow{D(K_s(\mathscr{J}^{(b)}), \pi)} & D(K_s(\mathscr{J}^{(b)}), i_s(M_2^{(b)})). \\
\downarrow \Psi & & \downarrow \Psi \\
D(K_s(\mathscr{J}), i_s(M_1)) & \xrightarrow{D(K_s(\mathscr{J}), \pi)} & D(K_s(\mathscr{J}), i_s(M))
\end{array}$$

By commutativity of the bottom square,

$$d' \circ \pi\Psi = d'\Psi \circ \pi = d\varphi\Psi.$$

Since Ψ is an isomorphism $d'o\pi = d\varphi$. By a previous result, there is a

$$\gamma \in I(K_s(\mathscr{J}^{(b)}), i_s(M_1^{(b)}))$$

such that

$$d' - \gamma = \delta\varphi, \delta \in D(\mathscr{J}^{(b)}, M_1^{(b)}).$$

Then

$$\delta o\pi\varphi = \delta\varphi o\pi = (d' - \gamma)\, o\pi = d'o\pi - \gamma o\pi = d\varphi - \gamma o\pi.$$

Hence

$$\delta o\pi\varphi \cong d\varphi(I(K_s(\mathscr{J}^{(b)}), i_s(M_2^{(b)})).$$

We have seen in Glassman [26],

$$(\delta o\pi - d)\,\varphi \in I(K_s(\mathscr{J}^{(b)}), i_s(M_2^{(b)}))$$

implies

$$\delta o\pi - d = \sum [R_a^{(b)} R_{m_2}^{(b)}], \{a\} \subseteq \mathscr{J}^{(b)}, \{m_2\} \subseteq M_2^{(b)}.$$

Then

$$\delta o\pi = d + \sum [R_a^{(b)} R_{m_2}^{(b)}].$$

Since π is an epimorphism, we can lift each m_2 in the sum above to $m_1 \in M_1$ yielding

$$\sum [R_a^{(b)} R_{m_1}^{(b)}]\, o\pi = \sum [R_a^{(b)} R_{m_2}^{(b)}].$$

Then

$$\left(\delta - \sum [R_a^{(b)} R_{m_1}^{(b)}]\right) o\pi = \delta o\pi - \sum [R_a^{(b)} R_{m_2}^{(b)}]\, o\pi = d.$$

We have so far succeeded in lifting d to a derivation

$$\delta - \sum [R_a^{(b)} R_{m_1}^{(b)}] \in D(\mathscr{J}^{(b)}, M_1^{(b)}).$$

An easy calculation shows that

$$\left(\delta - \sum [R_a^{(b)} R_{m_1}^{(b)}]\right) o\Phi = 0.$$

Hence

$$\delta - \sum [R_a^{(b)} R_{m_1}^{(b)}] \in J(\mathscr{J}^{(b)}, M_1^{(b)}) \qquad \text{and} \qquad J(\mathscr{J}^{(b)}, \ )$$

preserves epimorphisms. ∎

By returning to the construction of $M^{(b)}$ and the isomorphism of

$$\mathscr{H}_n(\mathscr{O}, \bar{R}) \simeq \mathscr{H}_n(\mathscr{O}, R)^{(b)},$$

it is easy to see that if (N, r) is an $(\mathscr{O}, r)$ bimodule, then

$$\Gamma(N, r)^{(b)} = (N_n \cap \mathscr{H}_n(E(\mathscr{O}, N), R))^{(b)}) \cong N_n \cap \mathscr{H}_n(E(\mathscr{O}, N), \bar{R}).$$

Then $\Gamma(N, r)^{(b)}$ is injective if and only if (N, r) is injective if and only if $\Gamma(N, r)$ is injective; that is, $M^{(b)}$ is an injective $\mathscr{J}^{(b)}$ bimodule if and only if M is an injective $\mathscr{J}$ bimodule.

COROLLARY 4′. *If $\bar{R}$ is any canonical involution, $\bar{M}$ an injective bimodule for $\mathscr{H}_n(\mathscr{O}, \bar{R})$, then*

$$H_{\mathscr{J}}^1(\mathscr{H}_n(\mathscr{O}, \bar{R}), \bar{M}) = 0.$$

Proof. Again, write

$$\mathscr{H}_n(\mathscr{O}, \bar{R}) = \mathscr{J}^{(b)}, \qquad \bar{M} = M^{(b)}.$$

We have seen that

$$D(\mathscr{J}^{(b)}, M^{(b)}) \xrightarrow{\Phi} \frac{D(K_s(\mathscr{J}), i_s(M))}{J(K_s(\mathscr{J}), i_s(M))}$$

is an epimorphism with kernal $J(\mathscr{J}^{(b)}, M^{(b)})$. Hence

$$\frac{D(\mathscr{J}^{(b)}, M^{(b)})}{J(\mathscr{J}^{(b)}, M^{(b)})} \simeq \frac{D(K_s(\mathscr{J}), i_s(M))}{J(K_s(\mathscr{J}), i_s(M))} \simeq \frac{D(\mathscr{J}, M)}{J(\mathscr{J}, M)}.$$

But $M^{(b)}$ injective implies M injective implies

$$\frac{D(\mathscr{J}, M)}{J(\mathscr{J}, M)} = 0$$

by Corollary 4. Hence

$$\frac{D(\mathscr{J}^{(b)}, M^{(b)})}{J(\mathscr{J}^{(b)}, M^{(b)})} = 0. \quad \blacksquare$$

From now on, R will denote any canonical involution.

LEMMA 15. $H^k(\mathscr{H}_n(\mathscr{O}, R), \Gamma(N, r)) \simeq H^k((\mathscr{O}, r), (N, r))$ *for all* $k \geqslant 2$.

Proof. For $k = 2$, Jacobson [15] has shown this. For $k > 2$, it will be sufficient to show that if

$$0 \to N_1 \to N_2 \to N_3 \to 0$$

is an exact sequence of $(\mathcal{A}, r)$ bimodules, then

$$0 \to \Gamma(N_1, r_1) \to \Gamma(N_2, r_2) \to \Gamma(N_3, r_3) \to 0$$

is exact, and conversely. But this is immediate from the definitions. If

$$\eta \in \mathrm{Hom}_{(U_1(\mathcal{A}),\bar{\sigma})}((N_1, r_1), (N_2, r_2)), \quad V = \ker \eta,$$

then

$$\ker \Gamma(\eta) \equiv V_n \cap \mathscr{H}_n(E(\mathcal{A}, V), R) = \Gamma(V, r|V)$$

and similarly for the cokernel. ∎

From this,

$$H^2(\mathscr{H}_n(\mathcal{A}, R), M) = 0$$

for M injective. As discussed in Glassman [26], choosing any finite set of generators for J, we can construct a $U(\mathscr{H}_n(\mathcal{A}, R))$-module C such that

$$H_J^k(\mathscr{H}_n(\mathcal{A}, R), \ )$$

is the k-th right derived functor of

$$\mathrm{Hom}_{U(\mathscr{H}_n(\mathcal{A},R))}(C, \ );$$

R any canonical involution. As was the case with involution, the module C above is not necessarily the simplest or most advantageous. The situation now indicates a simpler choice for C, a choice that relates to the cohomology of $(\mathcal{A}, r)$ and allows the extension of Corollary 3 to canonical involutions.

Recall that $H_I^0((\mathcal{A}, r), (N, r))$ was defined in Glassman [26] to be

$$\mathrm{Hom}_{(U_1(\mathcal{A}),\bar{\sigma})}((\mathcal{A}, -r), (N, r)).$$

Write $(\bar{\mathcal{A}}, -r)$ for the bimodule with involution $(\mathcal{A}, -r)$. Define

$$C_J = \bar{\mathcal{A}}_n \cap \mathscr{H}_n(E(\mathcal{A}, \bar{\mathcal{A}}), R).$$

From the equivalence of categories previously discussed,

$$\begin{aligned} H_J^0(\mathscr{H}_n(\mathcal{A}, R), \Gamma(N, r)) &\equiv \mathrm{Hom}_{U(\mathscr{H}_n(\mathcal{A},R))}(C_J, \Gamma(N, r)) \\ &\simeq \mathrm{Hom}_{(U_1(\mathcal{A}),\bar{\sigma})}((\mathcal{A}, -r), (N, r)) = H_I^0((\mathcal{A}, r), (N, r)). \end{aligned}$$

This, Corollary 4′, and Lemma 15, imply that

$$H_I^k((\mathcal{A}, r), (N, r)) \simeq H_J^k(\mathscr{H}_n(\mathcal{A}, R), \Gamma(N, r))$$

for all (N, r) and all $k \geqslant 0$, as discussed in Glassman [26].

COROLLARY 5. *If R is any canonical involution, (N, r) any $(\mathcal{A}, r)$ bimodule,*

$$H_J^1((\mathcal{A}, r), (N, r)) \simeq H_J^1(\mathcal{H}_n(\mathcal{A}, r), \Gamma(N, r)). \quad \blacksquare$$

COROLLARY 6. *If R is any canonical involution, $H_J^k(\mathcal{H}_n(\mathcal{A}, R), \ )$ is the k-th right derived functor of*

$$H_J^0(\mathcal{H}_n(\mathcal{A}, R), \ ) \equiv \mathrm{Hom}_{U(\mathcal{H}_n(\mathcal{A}, R))}(C_J, \ ). \quad \blacksquare$$

LEMMA 16. *If (N, r) is a projective bimodule for $(\mathcal{A}, r)$ in the category of bimodules with involution, then N is a projective $\mathcal{A}$ bimodule.*

Proof. Let (X, r) be a free bimodule with involution which projects onto (N, r). Then the standard proof shows that (N, r) is a direct summand (with involution) of (X, r). This implies that N is a direct summand of X. But if (X, r) is free on $\{x_i\}$, then X is free on $\{x_i, x_i r\}$. So N is a direct summand of a free bimodule, and is projective. $\blacksquare$

THEOREM 3. *C_J is a projective $\mathcal{H}_n(\mathcal{A}, R)$-bimodule if and only if $\mathcal{H}_n(\mathcal{A}, R)$ is separable.*

Proof. If $\mathcal{H}_n(\mathcal{A}, R)$ is separable, $U(\mathcal{H}_n(\mathcal{A}, R))$ is separable, so all finite dimensional $\mathcal{H}_n(\mathcal{A}, R)$-bimodules are projective. In particular, C_J is projective.

Suppose that C_J is projective. Then, as we did for injectives, we can show that projective bimodules for $\mathcal{H}_n(\mathcal{A}, R)$ correspond to projective bimodules for $(\mathcal{A}, r)$. Since

$$C_J = \bar{\mathcal{A}}_n \cap \mathcal{H}_n(E(\mathcal{A}, \bar{\mathcal{A}}), R),$$

$(\bar{\mathcal{A}}, -r)$ is projective with involution. By the previous lemma, $\mathcal{A}$ is a projective $\mathcal{A}$ bimodule. It is a standard result for finite dimensional associative algebras that $\mathcal{A}$ is separable if and only if $\mathcal{A}$ is projective as a bimodule over itself. Therefore $\mathcal{A}$ is separable. We need to show, now, that $\mathcal{A}$ separable implies $\mathcal{H}_n(\mathcal{A}, R)$ separable; and since

$$\mathcal{H}_n(\mathcal{A}_F, R_F) \simeq \mathcal{H}_n(\mathcal{A}, R)_F,$$

where F is a field extension of K, it will suffice to show that $\mathcal{A}$ semisimple implies $\mathcal{H}_n(\mathcal{A}, R)$ semisimple.

If $\mathcal{A}$ is semisimple, then so is $(\mathcal{A}, \sigma)$ for any involution σ. $(\mathcal{A}, \sigma)$ has summands which are either simple ideals of $\mathcal{A}$ or of the form $\mathcal{A}_1 \oplus \mathcal{A}_2$, $\mathcal{A}_i$ simple and $\mathcal{A}_2 \simeq \mathcal{A}_1$ under σ. Consider $(\mathcal{A}, r)$ as an $(\mathcal{A}, r)$ bimodule. Since the $\mathcal{H}_n(\mathcal{A}, R)$-bimodule $\mathcal{H}_n(\mathcal{A}, R)$ is isomorphic to $\mathcal{A}_n \cap \mathcal{H}_n(E(\mathcal{A}, \mathcal{A}), R)$, $\mathcal{H}_n(\mathcal{A}, R)$ is completely reducible as an $\mathcal{H}_n(\mathcal{A}, R)$-bimodule. Hence $\mathcal{H}_n(\mathcal{A}, R)$ is semisimple. $\blacksquare$

COROLLARY 7. *$\mathscr{H}_n(\mathscr{A}, R)$ is separable if and only if*

$$\frac{D(\mathscr{H}_n(\mathscr{A}, R), M)}{J(\mathscr{H}_n(\mathscr{A}, R), M)} = 0$$

for all bimodules M.

Proof.

$$H_J^1(\mathscr{H}_n(\mathscr{A}, R), M) = Ext^1_{U(\mathscr{H}_n(\mathscr{A}, R))}(C_J, M).$$

But

$$Ext^1_{U(\mathscr{H}_n(\mathscr{A}, R))}(C_J, M) = 0$$

for all M if and only if C_J is projective, and C_J is projective if and only if $\mathscr{H}_n(\mathscr{A}, R)$ is separable. ∎

COROLLARY 8. *If $(\mathscr{A}, r)$ is associative with involution and*

$$\frac{D((\mathscr{A}, r), (M, r))}{I((\mathscr{A}, r), (M, r))} = 0$$

for all bimodules with involution (M, r), then

$$\frac{D(\mathscr{A}, M)}{I(\mathscr{A}, M)} = 0$$

for all bimodules M.

Proof.

$$H_I^1((\mathscr{A}, r), (M, r)) = Ext^1_{(U_1(\mathscr{A}), \bar{\sigma})}((\mathscr{A}, -r), (M, r)).$$

But

$$Ext^1_{(U_1(\mathscr{A}), \bar{\sigma})}((\mathscr{A}, -r), (M, r)) = 0$$

for all (M, r) if and only if $(\mathscr{A}, -r)$ is projective with involution. By Lemma 16, this occurs only if $\mathscr{A}$ is a projective $\mathscr{A}$ bimodule, implying $\mathscr{A}$ separable and all bimodules are injective. ∎

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